

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 21

LARGEST NUMBER IN AN ARRAY

AIM: To write and execute an 8085 assembly language program to find largest number in an array and store result in memory.

ALGORITHM

1. Store the array elements in consecutive memory locations starting from **2050H**.
2. Load the number of elements into register **C**.
3. Assume the first element as the largest number.
4. Compare the largest number with the next array element.
5. If the next element is greater, update the largest number.
6. Repeat the comparison for all elements.
7. Store the largest number in **2055H**.
8. Stop the program.

CODE:

```
MVI C,05H
```

```
LXI H,2050H
```

```
MOV A,M
```

```
INX H
```

```
DCR C
```

```
LOOP: CMP M
```

```
    JNC NEXT
```

```
    MOV A,M
```

```
NEXT: INX H
```

```
    DCR C
```

```
    JNZ LOOP
```

STA 2055H

HLT

OUTPUT:

GNUSim8085 - 8085 Microprocessor Simulator

FileResetAssemblerDebugHelp

Registers

A	00	S	0
BC	00 00	Z	1
DE	00 00	AC	0
HL	20 55	P	1
PSW	00 00	C	0
PC	42 16		
SP	FF FF		
Int-Reg	00		

Flag

S	0
Z	1
AC	0
P	1
C	0

Decimal - Hex Conversion

Decimal	Hex
0	0

To HexTo Dec

I/O Ports

0	-	00
---	---	----

Update Port Value

Memory

2054	-	06
------	---	----

Update Memory

Load me at

```
1 MVI C,05H
2 LXI H,2050H
3 MOV A,M
4 INX H
5 DCR C
6
7 LOOP: CMP M
8       JNC NEXT
9       MOV A,M
10
11 NEXT: INX H
12       DCR C
13       JNZ LOOP
14
15 STA 2055H
16 HLT
```

DataStackKeyPadMemoryI/O Ports

Start2050OK

Address (Hex)	Address	Data
0802	2050	12
0803	2051	20
0804	2052	25
0805	2053	8
0806	2054	6
0807	2055	25
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0

Line NoAssembler Message

0Program assembled successfully

Simulator: Idle

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